

Convergencia casi segura

Definición 1 (Convergencia casi segura). Sea $(X_n)_{n \in \mathbb{N}}$ una sucesión de variables aleatorias, X_n converge casi seguro a X

$$\iff \exists A \in \Sigma : \mathbb{P}(A) = 0 \wedge \forall \omega \in A^c : \lim_{n \rightarrow \infty} X_n(\omega) = X(\omega) \iff X_n \xrightarrow[n \rightarrow \infty]{\text{c.s.}} X.$$

Es decir, $X_n \xrightarrow[n \rightarrow \infty]{\text{c.s.}} X \iff X_n$ converge puntualmente a X en casi todo $\omega \in \Omega$.

Referenciado en

- prop-borel-cantelli-iii
- convergencia-casi-segura-imp-probabilidad
- ley-fuerte-grandes-numeros
- obs-convergencia-casi-segura-carac
- convergencia-probabilidad-imp-subsucesion-casi-segura
- teo-convergencia-martingalas

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